

Eurostat Data Sources

Germany's Tourism Competitiveness in the European Union

Ironhack Data Analytics Final Project | Jan Noel Vero | 2021–2026 YTD

Primary Data Source

The primary source for the project is Eurostat, the statistical office of the European Union. Official Eurostat tourism accommodation datasets were used for foreign arrivals, overnight stays, source-market analysis, seasonality, EU benchmarking and 2026 YTD analysis. Eurostat balance-of-payments travel data were used for the economic context.

Eurostat Datasets Used

1. Monthly nights spent at tourist accommodation

Dataset code: tour_occ_nim

Project purpose: Used for monthly foreign overnight stays, seasonality, historical analysis, EU comparison and 2026 YTD analysis.

Official Eurostat link: https://ec.europa.eu/eurostat/databrowser/view/tour_occ_nim/default/table?lang=en

2. Monthly arrivals at tourist accommodation

Dataset code: tour_occ_arm

Project purpose: Used for monthly foreign arrivals, EU benchmarking and calculation/comparison of average length of stay.

Official Eurostat link: https://ec.europa.eu/eurostat/databrowser/view/tour_occ_arm/default/table?lang=en

3. Annual nights by country of residence

Dataset code: tour_occ_ninraw

Project purpose: Used for annual foreign overnight stays by source market and Germany's source-market strategy analysis.

Official Eurostat link:

https://ec.europa.eu/eurostat/databrowser/view/tour_occ_ninraw/default/table?lang=en

4. Annual arrivals by country of residence

Dataset code: tour_occ_arnraw

Project purpose: Used for annual arrivals by visitor origin/source market.

Official Eurostat link:

https://ec.europa.eu/eurostat/databrowser/view/tour_occ_armraw/default/table?lang=en

5. International trade in services / Balance of Payments

Dataset code: bop_its6_det

Project purpose: Used for Germany's travel receipts, travel expenditure and travel balance economic context.

Official Eurostat link: https://ec.europa.eu/eurostat/databrowser/view/bop_its6_det/default/table?lang=en

Eurostat API

The project used a reproducible Python data-collection workflow. Eurostat provides programmatic access to datasets through its Statistics API, allowing dataset codes to be queried and processed in Python.

Eurostat API documentation: <https://ec.europa.eu/eurostat/web/user-guides/data-browser/api-data-access/api-getting-started/api>

How the Sources Map to the Project

Project Analysis	Eurostat Dataset	Main Indicator
EU competitiveness benchmark	tour_occ_arm / tour_occ_nim	Foreign arrivals and overnight stays
Average length of stay	tour_occ_arm / tour_occ_nim	Nights ÷ arrivals
Seasonality	tour_occ_nim	Monthly foreign nights
Source-market analysis	tour_occ_ninraw / tour_occ_armraw	Annual nights and arrivals by residence
2026 YTD analysis	tour_occ_arm / tour_occ_nim	Jan–Jun arrivals, nights and LOS
Economic context	bop_its6_det	Travel receipts, expenditure and balance

Data Handling and Interpretation

- Raw Eurostat data were retained unchanged before cleaning and transformation.
- Missing or confidential observations were not automatically interpreted as zero.
- Eurostat status information, including provisional/confidential indicators where relevant, was retained during processing.
- Historical annual EU comparisons required complete monthly observations under the project's completeness rules.
- 2026 was treated as year-to-date (January–June) and compared with equivalent periods rather than full-year results.
- Regional, EU and world aggregates were excluded from the eligible source-market analysis to avoid double counting.
- Balance-of-payments travel statistics have a broader scope than accommodation statistics and were therefore used as economic context rather than treated as directly interchangeable measures.

Suggested Ironhack Presentation Statement

My primary data source was Eurostat, the statistical office of the European Union. I used official accommodation datasets for monthly and annual foreign arrivals and overnight stays, including country-of-residence data for source-market analysis. I also used Eurostat's balance-of-payments travel dataset for the economic context. The data were collected through a reproducible workflow, preserved in raw form, cleaned under explicit data-quality rules, and then transformed for statistical analysis, SQL and Tableau.

Project Evidence

The data-collection process is documented in notebooks/01_data_mining_collection.ipynb. Cleaning and validation rules are documented in notebooks/02_data_cleaning.ipynb, while the raw, clean and processed data are separated in the project's data directory.